Answer **all** the questions in the spaces provided.

For Examiner's Use

1 A spring is attached at one end to a fixed point and hangs vertically with a cube attached to the other end. The cube is initially held so that the spring has zero extension, as shown in Fig. 1.1.

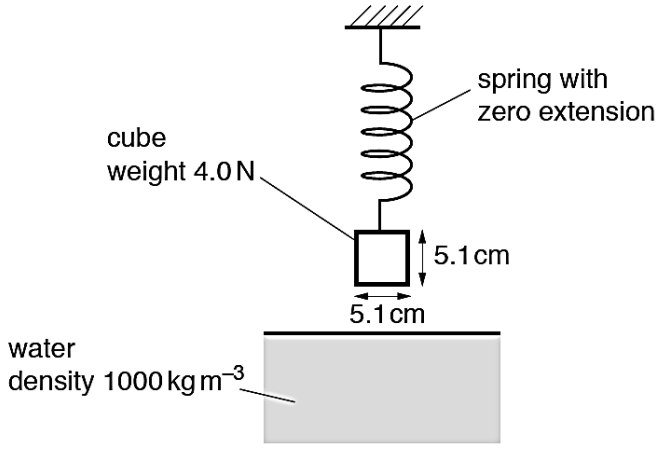

Fig. 1.1

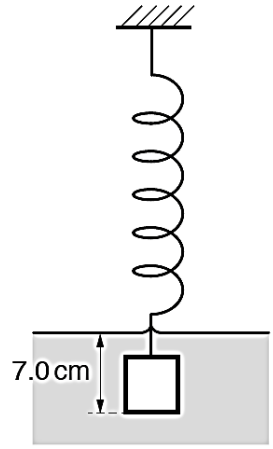

Fig. 1.2

The cube has weight 4.0 N and sides of length 5.1 cm. The cube is released and sinks into water as the spring extends. The cube reaches equilibrium with its base at a depth of 7.0 cm below the water surface, as shown in Fig. 1.2.
The density of water is 1000 kg m^{-3}.

(a) **(i)** Calculate the difference in the pressure exerted by the water on the bottom face and on the top face of the cube.

difference in pressure = ……. Pa [2]

(ii) Use your answer in **(a)(i)** to show that the upthrust on the cube is 1.3 N. Explain your working.

[2]

(iii) Calculate the force exerted on the spring by the cube when it is in equilibrium in the water.

force = ……. N [2]

For Examiner's Use

(iv) The spring obeys Hooke's law and has a spring constant of 30 N m^{-1}.

Determine the initial height above the water surface of the base of the cube before it was released.

height above surface = ……. cm [2]

(b) The cube in the water is released from the spring.

(i) State Newton's second law of motion.

……

……

…… [2]

(ii) Determine the initial acceleration of the cube.

acceleration = ……. m s^{-2} [2]

(iii) Describe and explain the variation, if any, of the acceleration of the cube as it sinks further into the water.

……

……

…… [2]